



Lecture #7: Dynamic Games

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Math 190: Mathematics of Sustainability, Emory University

February 9, 2026



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Dynamic Games





Dynamic Games

- Now there may be an order in which the players make their moves.
- we need to formalize this idea
- use our insights from static games.





Sequential Move Hanging with Friends

Recall the "hanging with friends" game, where two friends want to hang out but they don't know which venue the other will go to. We can represent it as a table (Tab: 5)

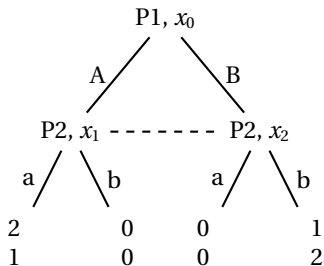
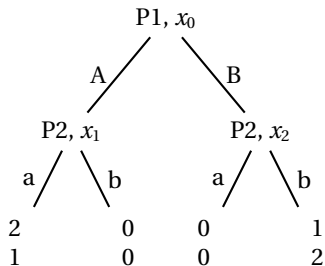
		Player 2	
		<i>A</i>	<i>B</i>
Player 1	<i>A</i>	2,1	0,0
	<i>B</i>	0,0	1,2

However, if we want dynamics, we can represent it as a tree.





Information sets





Who Wins in sequential games?

For 10 minutes: think about hanging with friends as a sequential game. Compare this to Rock, Paper, Scissors as a sequential game. Who is at an advantage.





Types of Information

Definition

Dynamic games of complete information are called

- games of *perfect information* if every information set is a singleton
- otherwise called *imperfect information*, e.g the information set is:

Definition (Extensive form Game)

An *extensive form game*

- 1 set of players,
- 2 payoff for all players as a function of actions
- 3 order of the moves
- 4 actions of players when they can move
- 5 knowledge of player when they can move (modeled using “information sets”)
- 6 Nature can be a player
- 7 structure of the game is common knowledge to all players



Mixed Strategies in Sequential Games

Example (Mixed Strategy example)

$$\sigma_2 = \begin{cases} \sigma_2(aa) = \frac{1}{10} \\ \sigma_2(ab) = \frac{3}{5} \\ \sigma_2(ba) = \frac{3}{10} \\ \sigma_2(bb) = 0 \end{cases}$$

This means if player 2 is at node x_1 , meaning player 1 played A , then player 2 will play a with probability $\frac{1}{10} + \frac{3}{5}$. And player 2 will play f with probability $\frac{3}{10} + 0$.





Behavioral and Mixed Strategies

Definition

Behavioral Strategy A *behavioral strategy* specifies a probability distribution over $A_i(h_i)$ for each information set $h_i \in H_i$

Remark

If a player has multiple turns, then these distributions are assumed to be independent, and for good reason.

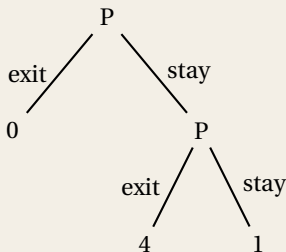




Behavioral and Mixed Strategies (cont.)

Example

The concepts of behavioral and mixed strategies aren't the same, as illustrated below.



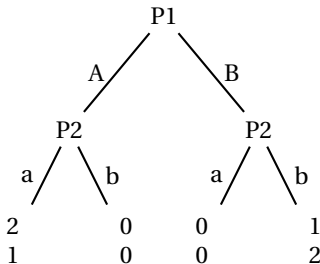
A randomized strategy will never receive a payoff of 4, because player 1 will take the same action in both nodes. However, a behavioral strategy flips a fresh coin at each node.

Behavioral and mixed strategies are the same in perfect information games.



Analyzing the Games

Let's start analyzing this game! The strategies for P1 are A , B , and the strategies for P2 are deterministic plan of action.





Sequential Games as Normal Form Games

- ① Players: $N = 2$
- ② Strategies:
 - $s_1 = \{A, B\}$
 - $s_2 = \{aa, ab, ba, bb\}$
- ③ payoffs:

		Player 2			
		<i>aa</i>	<i>ab</i>	<i>ba</i>	<i>bb</i>
Player 1	<i>A</i>	2,1	2,1	0,0	0,0
	<i>B</i>	0,0	1,2	0,0	1,2





Finding the True Nash Equilibria

